

Project Orion - Technical Specification v0.3

Overview

Orion is an internal document search service. It ingests PDFs from the shared drive, embeds them, and exposes a question-answering endpoint to staff.

Status: draft. Owner: Platform team. Target release: Q3.

Functional Requirements

- FR-1 The system shall ingest PDF, DOCX and CSV files up to 50 MB each.
- FR-2 The system shall return an answer within 5 seconds at the 95th percentile.
- FR-3 Every answer shall cite the source document and page number.
- FR-4 The system shall refuse to answer when retrieval confidence is below 0.35.
- FR-5 Administrators shall be able to delete a document and its vectors together.

Non-Functional Requirements

- NFR-1 Availability target is 99.5 percent during business hours.
- NFR-2 All documents remain within the ap-south-1 region.
- NFR-3 No document content may be sent to third-party model providers without an executed data processing agreement.

Architecture Decisions

- ADR-01 We use a flat FAISS index rather than an approximate index because the corpus is under 100,000 chunks and exact search keeps recall predictable.
- ADR-02 Chunk size is fixed at 1000 characters with 150 characters of overlap. Semantic chunking was evaluated and rejected for v1 on latency grounds.
- ADR-03 Embeddings are generated locally with a sentence-transformer model so no document text leaves the network during indexing.

Out of Scope for v1

Optical character recognition for scanned documents. Multi-language support beyond English. Per-user access control; v1 assumes all staff may read all indexed documents.

Open Questions

- Q1 How should the system handle documents that are superseded but not deleted?
- Q2 Do we need an audit log of every query, and for how long must it be kept?